

NileCAM81_CUOAGX

Application Note:

Trigger Mode



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e-con Systems

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Contents

INTRODUCTION TO NILECAM81_CUOAGX	4
DESCRIPTION	5
TRIGGER MODE	7
EXTERNAL TRIGGER SETUP FOR NILECAM81_CUOAGX WITH JETSON ORIN DEVELOPER KIT	7
ENABLING EXTERNAL TRIGGER	10
PWM CONFIGURATION STEPS ON NVIDIA BOARD	10
PWM CONFIGURATION STEPS USING SYSFS ENTRY	11

Introduction to NileCAM81_CUOAGX

NileCAM81_CUOAGX is a family of NileCAM camera module from e-con Systems. It is provided with a high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81_CUOAGX offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81_CUOAGX fast (low latency), flexible and yet reliable.

NileCAM81_CUOAGX uses Maxim's Gigabit Multimedia Serial Link (GMSL) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81_CUOAGX embeds the Maxim's serializer which serializes the high-speed image data and transmits over coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81_CUOAGX serializer recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81_CUOAGX board has been designed and developed by e-con Systems for the NVIDIA® Jetson AGX Orin™/Xavier™ development kit. NileCAM81_CUOAGX is an 8 MP GMSL camera contains a 1/1.7" AR0821 CMOS image sensor from onsemi™ along with on-board ISP. The coaxial cable is connected to camera module enclosure through a rugged FAKRA connector. NileCAM81_CUOAGX is interfaced to the J509 connector on the Orin™/Xavier™ using the NileCAM81_CUOAGX deserializer board.

The NVIDIA® Jetson Orin™/Xavier™ developer kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope.

e-con Systems provides the sample applications that demonstrate the features of this camera. However, this camera can also be utilized by any V4L2 application.

This document describes external trigger setup to interface NileCAM81_CUOAGX board on Jetson Orin™/Xavier™ development kit.

Description

This Application Note concentrates on the working of trigger mode in NileCAM81_CUOAGX. Many imaging applications commonly requires the image sensor to capture an image after the triggering action. This triggering action can be the passing of an object on a conveyor belt, the flash of a strobe light, or the click of a button.

The NileCAM81_CUOAGX board is designed to interface one or six 8MP camera modules with the Jetson Orin™/Xavier™ development kit.

NileCAM81_CUOAGX is a multi-board solution, which has three boards as follows:

- e-CAM82_CUMI0821_MOD_H02R1 (Camera module)
- e-CAM22_CUMI_SER (Serializer Board)
- e-CAM_CUOAGX_DESER_6H01R1 (Deserializer Board)

The front and rear views of NileCAM81 camera module and deserializer boards are shown in following figures.

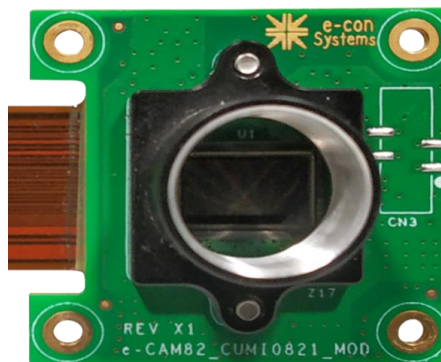


Figure 1: Front View of NileCAM81 Camera Board



Figure 2: Rear View of NileCAM81 Serializer Board



Figure 3: Front view of NileCAM81_CUOAGX Deserializer Board



Figure 4: Rear view of NileCAM81_CUOAGX Deserializer Board

The camera board is based on AR0821, 1/1.7" CMOS image sensor from onsemi™ along with ISP. The camera is available with M12 lens holder allowing customers to choose the lens as per their requirements. The serializer board is based on Maxim GMSL serializer and has all the circuitry required for bidirectional data communication, power and data separation and so on. FAKRA connectors are used for the coaxial cable interface. The deserializer board needs external power supply, which supplies the power to the Serializer board over coaxial cable using FAKRA connector. The deserializer board has six FAKRA connectors, through which you can connect six cameras at the same time.

NileCAM81_CUOAGX is a three-board solution. The camera module has two boards, module board and serializer board, each of size as 30 mm x 30 mm. The dimension of deserializer board is 75 mm x 55 mm. NileCAM81_CUOAGX is available with both 3m and 15m coaxial cable options for power over coax applications.

Trigger Mode

NileCAM81_CUOAGX camera can synchronize the exposure (or integration) of the camera pixels to an Internal or external trigger pulse that can be given through the GPIO connector of the camera. By default NileCAM81_CUOAGX comes with Internal trigger setup. Below section describes the external trigger setup.

External Trigger Setup for NileCAM81_CUOAGX with Jetson Orin Developer Kit

1. Connect the 1 to 1 mating cable with part number (A06SR06SR30K152B) to the CN12 connector of Deserializer.

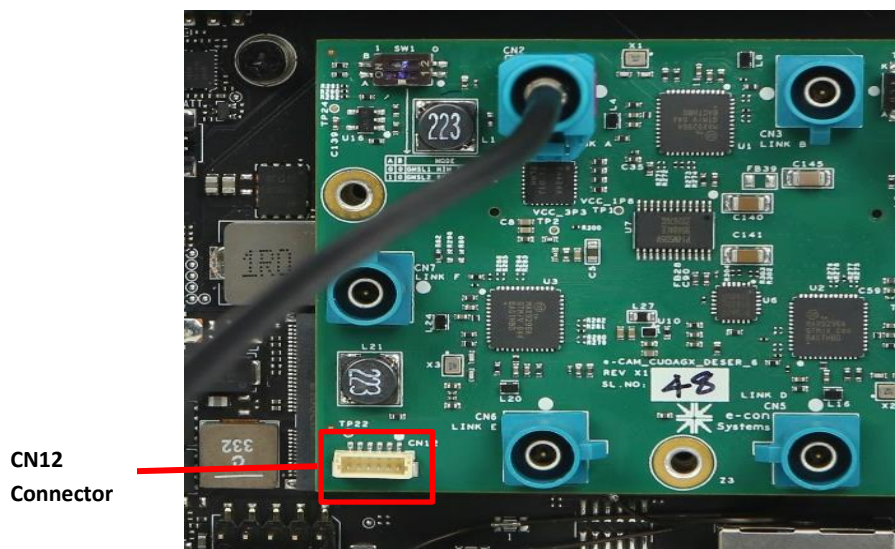


Figure 5: CN12 connector location of NileCAM81_CUOAGX Deserializer Board

2. Crimp the CN12 mating cable with female head connector for pin 2 (for trigger) and pin 6 (for ground) which should be connected to the pin 15 (for trigger) and 39 pin (for ground) of Jetson Orin/Xavier platform board's 40 pin header with help of male to female connecting wire.

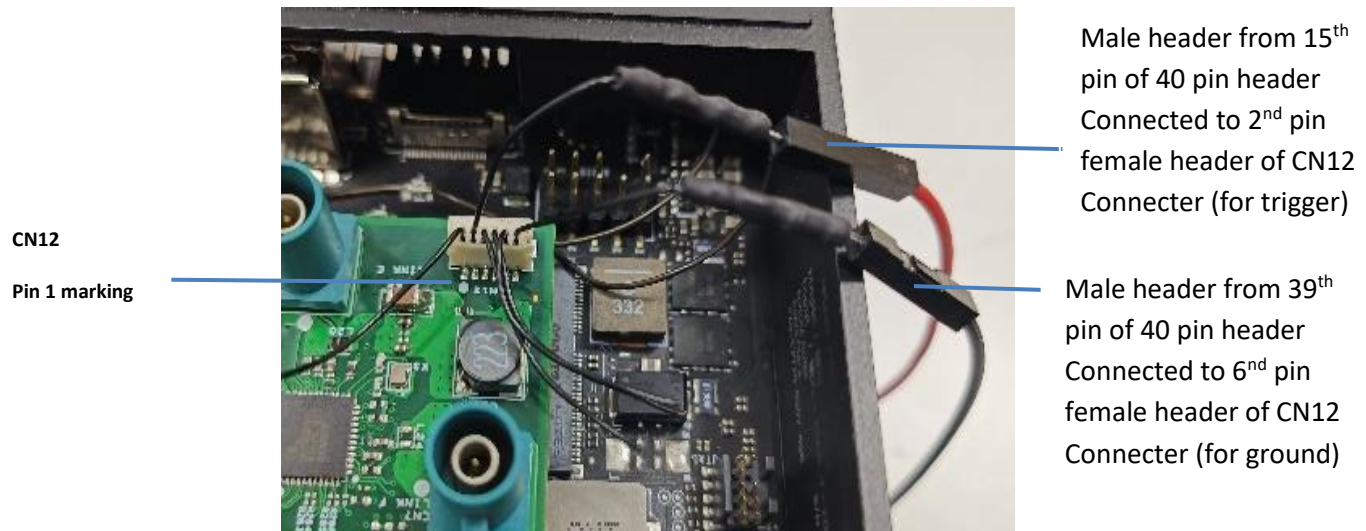


Figure 6: Connected 1 to 1 mating cable to CN12 Connector

3. Connect Pin 15 in Jetson board's 40 pin header to Deserailizer's External trigger connector CN12 with pin number 2 with the help of mating cable for PWM input and Connect Pin 39 in Jetson board's 40 pin header to Deserailizer's External trigger connector CN12 with pin number 6 with the help of mating cable for Ground.

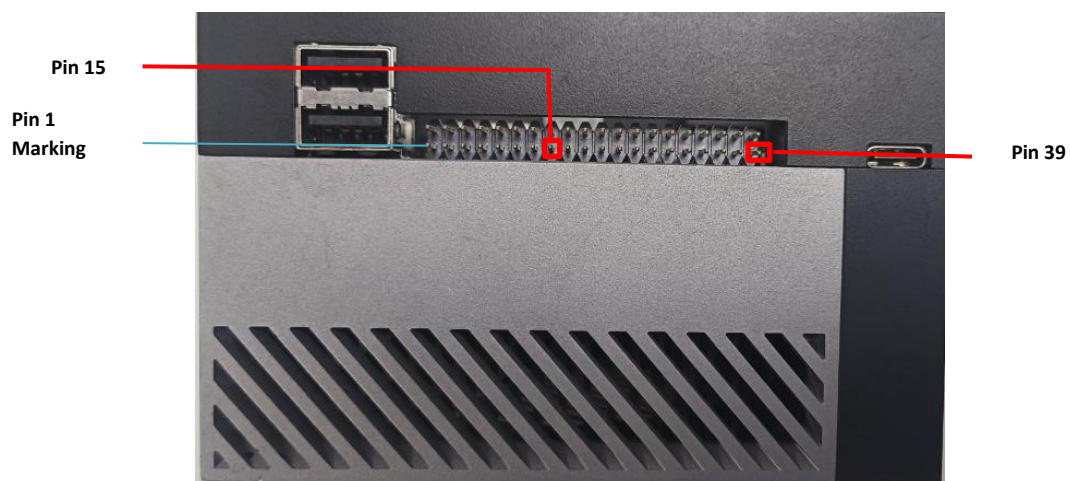


Figure 7: Pin 15 and pin 39 location of Jetson Orin™/Xavier™

Female header
Connected to 15th
pin of 40 pin
header (for trigger)



Female header
Connected to 39th
pin of 40 pin header
(for ground)

Figure 8: connection of Pin 15 and pin 39 location of Jetson Orin™/Xavier™

Caution: Make sure the trigger and ground pins from Jetson platform board is connected to the respective pins on the connector CN12. Otherwise it may cause the damage to board.

Enabling External Trigger

This section describes enabling external trigger with Jetson Orin™ / Xavier™ development Kit.

PWM Configuration Steps on NVIDIA Board

1. Run the Following Script:

```
sudo /opt/nvidia/jetson-io/jetson-io.py
```

You will see the below window:

```

===== Jetson Expansion Header Tool =====

3.3V ( 1) ( 2) 5V
i2c9 ( 3) ( 4) 5V
i2c9 ( 5) ( 6) GND
unused ( 7) ( 8) uarta
GND ( 9) (10) uarta
unused (11) (12) unused
unused (13) (14) GND
unused (15) (16) unused
3.3V (17) (18) unused
unused (19) (20) GND
unused (21) (22) unused
unused (23) (24) unused
GND (25) (26) unused
i2c2 (27) (28) i2c2
unused (29) (30) GND
unused (31) (32) unused
unused (33) (34) GND
unused (35) (36) unused
unused (37) (38) unused
GND (39) (40) unused

Select one of the following options:
Configure Jetson for compatible hardware
Configure 40-pin expansion header
Exit
  
```

2. Choose "**configure 40-pin expansion header**"

The Below window will be appeared.

```

===== Jetson Expansion Header Tool =====

Select desired functions (for pins):

[ ] aud_mclk      (7)
[ ] extperiph3_clk (29)
[ ] extperiph4_clk (31)
[ ] i2s5          (12,35,38,40)
[ ] pwm1          (33)
[ ] pwm5          (15,32,33)
[*] pwm8          (32)
[ ] spi1          (19,21,23,24,26)
[ ] spi3          (13,16,18,22,37)
[ ] uarta-cts/rts (11,36)

Back
  
```

3. Choose "**pwm1**" to generate a pulse (Here 33rd pin is configured for pwm1).

4. Then click "Back".

```

===== Jetson Expansion Header Tool =====

      3.3V ( 1) ( 2) 5V
      i2c9 ( 3) ( 4) 5V
      i2c9 ( 5) ( 6) GND
      unused ( 7) ( 8) uarta
      GND ( 9) (10) uarta
      unused (11) (12) unused
      unused (13) (14) GND
      unused (15) (16) unused
      3.3V (17) (18) unused
      unused (19) (20) GND
      unused (21) (22) unused
      unused (23) (24) unused
      GND (25) (26) unused
      i2c2 (27) (28) i2c2
      unused (29) (30) GND
      unused (31) (32) gp
      unused (33) (34) GND
      unused (35) (36) unused
      unused (37) (38) unused
      GND (39) (40) unused

Select one of the following options:
Save and reboot to reconfigure pins
Save and exit without rebooting
Export as Device-Tree Overlay
Discard pin changes
Exit

```

5. Then Click on "Save and reboot to reconfigure pins."

6. After restart, **Run the below script** to get the pwm on the pwm1 pin.

Note: Make sure the Pin-1 and pin-2 of J514 connector is shorted.

PWM Configuration steps using sysfs entry

Script to Generate 15HZ pulse on pwm1:

1. echo 0 > /sys/class/pwm/pwmchip0/export
2. echo 66666666 > /sys/class/pwm/pwmchip0/pwm0/period
3. echo 99999999 > /sys/class/pwm/pwmchip0/pwm0/duty_cycle
4. echo 1 > /sys/class/pwm/pwmchip0/pwm0/enable

Use oscilloscope to check for the square wave.

PWM Calculaton:

1) Period Calculation:

$$\text{period_value_in_ns} = (1/\text{Frequency}) * 1000000000$$

2) Duty Cycle Calculation

$$\text{duty_cycle_value_in_ns} = \text{period_value_in_ns} / (100 / \% \text{ of duty cycle})$$

Example Calculation:

1. For 30Hz with 50% Duty Cycle,

$$\text{period_in_ns} = (1 / 30) \times 1000000000 = 33333333$$

$$\text{duty_cycle_in_ns} = 33333333 / (100 / 50) = 16666666$$

Revision History

Rev	Date	Description	Author
1.0	10-Feb-2023	Initial Draft	Camera Dev Team